Renewable Energy Distribution Network Using Kruskal's Algorithm

1. Aim

To develop a renewable energy distribution graph consisting of power plants, substations, and weighted transmission links, and apply Kruskal's algorithm to construct a minimum-cost connected network.

2. Problem Statement

Renewable energy sources such as solar and wind power plants need to supply electricity to different substations through transmission lines. Constructing these transmission lines involves installation and maintenance costs.

The problem is to represent the power plants and substations as vertices and the transmission lines as weighted edges in a graph. Kruskal's algorithm is used to select the minimum-cost transmission links while ensuring that all locations are connected without forming cycles.

3. Objectives

- To represent the renewable energy distribution system using a weighted graph.
- To apply Kruskal's algorithm to find the Minimum Spanning Tree (MST).
- To minimize the total cost of transmission line installation.
- To connect all power plants and substations efficiently.
- To avoid unnecessary transmission links and cycles.
- To understand the practical application of graph algorithms in power distribution systems.

4. Advantages

- **Minimum Cost:** Reduces the total cost of selected transmission links.
- **Efficient Connectivity:** Connects all vertices using the minimum required number of edges.
- **Cycle Prevention:** Avoids unnecessary loops in the network.
- **Easy Implementation:** Kruskal's algorithm is straightforward to implement using C++.
- **Better Resource Utilization:** Helps in planning the use of transmission infrastructure.
- **Scalability:** Can be applied to larger networks with more power plants and substations.

5. Disadvantages

- **No Backup Routes:** The minimum spanning tree does not provide alternative paths if a transmission line fails.
- **Static Costs:** The algorithm may not account for changing installation or maintenance costs.
- **No Power Flow Analysis:** It does not calculate voltage drops, current, or power losses.
- **Limited Practical Factors:** It does not automatically consider geographical obstacles or environmental restrictions.
- **Single Objective:** It minimizes total edge weight but does not directly optimize reliability, capacity, or energy efficiency.
- **Network Failure Risk:** A failure in a selected edge can disconnect parts of the network.

6. Future Scope

- Integrate real-time data from solar and wind power plants.
- Include transmission losses and power capacity in network planning.
- Use smart grid technology for real-time monitoring and control.
- Develop a graphical user interface to display the network and selected links.
- Include backup transmission routes to improve reliability.
- Use AI and optimization techniques to support dynamic energy distribution planning.

7. Conclusion

Kruskal’s algorithm provides an effective method for designing a minimum-cost renewable energy distribution network. By selecting the lowest-weight transmission links without forming cycles, it connects all power plants and substations with minimum total edge cost. Although it does not directly consider power losses or backup routes, it provides a useful foundation for cost-effective network planning and can be enhanced for future smart grid applications.